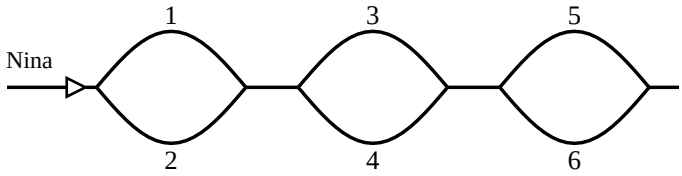
# Preview: set-03/by_level/Grade_03-04/y2007_q01 (v2)

Routes through forks: one branch from every fork; number of routes = product of choices

***Part A: Each question is worth 3 points***

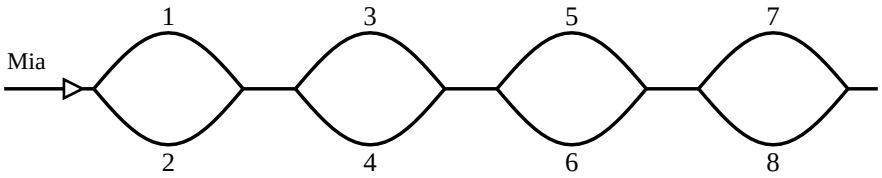
**1.** Nina walks along the road from left to right and writes down every number on the way.




Which group of numbers could Nina have written?

***A.*** 1, 2 and 4 ***B.*** 1, 4 and 5 ***C.*** 1, 3 and 4 ***D.*** 4, 5 and 6 ***E.*** 3, 4 and 5

**2.** Mia walks along the road from left to right and writes down every number on the way.

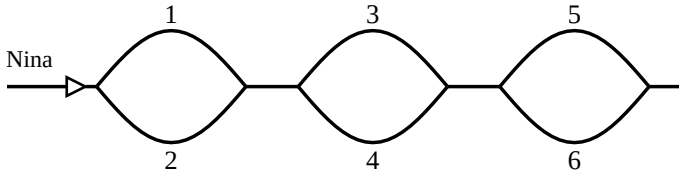



Which group of numbers could Mia have written?

***A.*** 1, 2, 4 and 8 ***B.*** 1, 3, 4 and 5 ***C.*** 1, 4, 7 and 8

***D.*** 1, 3, 4 and 8 ***E.*** 1, 4, 5 and 8

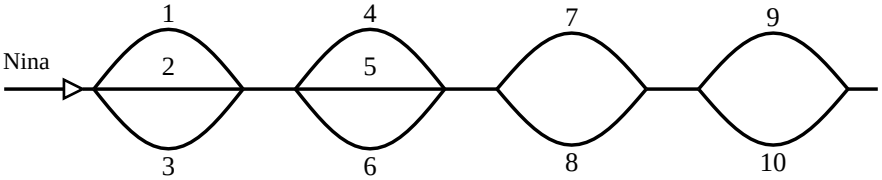
**3.** Nina walks along the road from left to right and writes down every number on the way.




How many different groups of numbers can Nina write?

***A.*** 10 ***B.*** 9 ***C.*** 8 ***D.*** 6 ***E.*** 3

**4.** Nina walks along the road from left to right and writes down every number on the way.

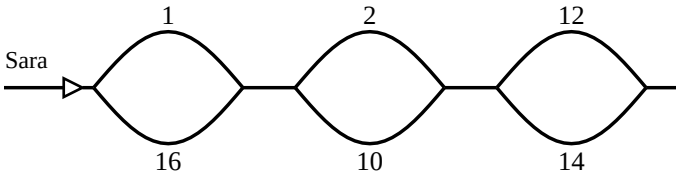



How many different groups of numbers can Nina write?

***A.*** 45 ***B.*** 43 ***C.*** 38 ***D.*** 36 ***E.*** 18

***Part B: Each question is worth 4 points***

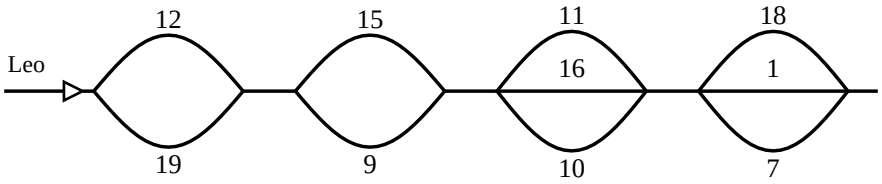
**5.** Sara walks along the road from left to right and adds up the numbers along the way.




What is the biggest total Sara can get?

***A.*** 40 ***B.*** 42 ***C.*** 44 ***D.*** 49 ***E.*** 55

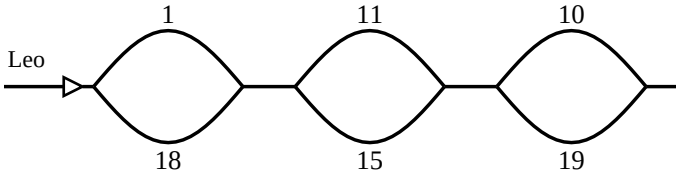
**6.** Leo walks along the road from left to right and adds up the numbers along the way.




What is the biggest total Leo can get?

***A.*** 68 ***B.*** 67 ***C.*** 54 ***D.*** 52 ***E.*** 32

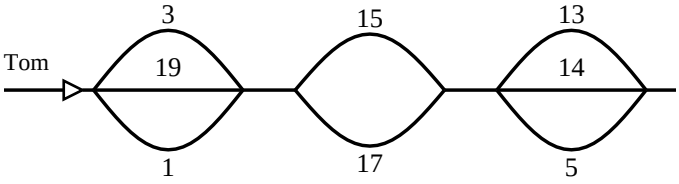
**7.** Leo walks along the road from left to right and adds up the numbers along the way.




Which total can Leo **not** get?

***A.*** 22 ***B.*** 29 ***C.*** 31 ***D.*** 39 ***E.*** 48

**8.** Tom walks along the road from left to right and adds up the numbers along the way.

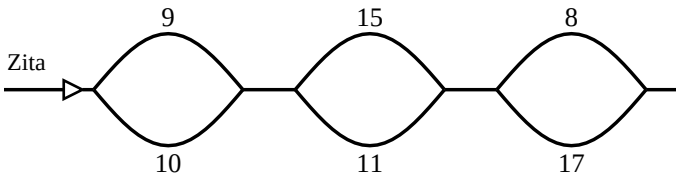



Which total can Tom **not** get?

***A.*** 23 ***B.*** 25 ***C.*** 28 ***D.*** 30 ***E.*** 49

***Part C: Each question is worth 5 points***

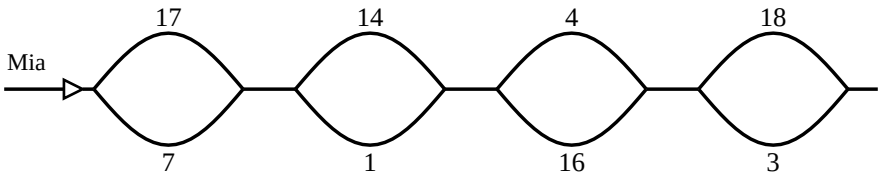
**9.** Zita walks along the road from left to right and adds up the numbers along the way. The total is 29.




Which number did Zita see at the second fork?

***A.*** 8 ***B.*** 9 ***C.*** 10 ***D.*** 11 ***E.*** 15

**10.** Mia walks along the road from left to right and adds up the numbers along the way. The total is 28.




Which number did Mia see at the second fork?

***A.*** 3 ***B.*** 4 ***C.*** 7 ***D.*** 14 ***E.*** 16

# Answer Key

| | | | | | |
|---|---|---|---|---|---|
| 1. **B** | 2. **E** | 3. **C** | 4. **D** | 5. **A** | 6. **A** |
| 7. **B** | 8. **C** | 9. **D** | 10. **D** | | |

### *Solutions*

**1. B** (1, 4 and 5) — At every fork Nina takes exactly one branch, so there must be one number from each fork: 1, 4 and 5 works. Each of the others has two numbers from the same fork. [set-03/by_level/Grade_03-04/2007 Q1 · which_group]

**2. E** (1, 4, 5 and 8) — At every fork Mia takes exactly one branch, so there must be one number from each fork: 1, 4, 5 and 8 works. Each of the others has two numbers from the same fork. [set-03/by_level/Grade_03-04/2007 Q1 · which_group]

**3. C** (8) — The forks have 2, 2, 2 branches; every choice can go with every other: $2 \times 2 \times 2 = 8$. [set-03/by_level/Grade_03-04/2007 Q1 · how_many_routes]

**4. D** (36) — The forks have 3, 3, 2, 2 branches; every choice can go with every other: $3 \times 3 \times 2 \times 2 = 36$. [set-03/by_level/Grade_03-04/2007 Q1 · how_many_routes]

**5. A** (40) — Take the bigger number at every fork: $16 + 10 + 14 = 40$. [set-03/by_level/Grade_03-04/2007 Q1 · largest_sum]

**6. A** (68) — Take the bigger number at every fork: $19 + 15 + 16 + 18 = 68$. [set-03/by_level/Grade_03-04/2007 Q1 · largest_sum]

**7. B** (29) — $22 = 1 + 11 + 10$; $31 = 1 + 11 + 19$; $39 = 18 + 11 + 10$; $48 = 18 + 11 + 19$. No route gives 29. [set-03/by_level/Grade_03-04/2007 Q1 · impossible_total]

**8. C** (28) — $23 = 3 + 15 + 5$; $25 = 3 + 17 + 5$; $30 = 1 + 15 + 14$; $49 = 19 + 17 + 13$. No route gives 28. [set-03/by_level/Grade_03-04/2007 Q1 · impossible_total]

**9. D** (11) — Only one route gives 29: $10 + 11 + 8 = 29$. At the second fork: 11. [set-03/by_level/Grade_03-04/2007 Q1 · which_branch]

**10. D** (14) — Only one route gives 28: $7 + 14 + 4 + 3 = 28$. At the second fork: 14. [set-03/by_level/Grade_03-04/2007 Q1 · which_branch]